

FRMCS Reference Specification — Migration Requirements

The FRMCS system shall provide continuous network coverage along all main railway lines, including tunnels and remote sections. The system shall support a minimum data throughput of 10 Mbps per train for critical applications. Emergency communication channels shall be established within two seconds of initiation under all network conditions.

The system shall authenticate all users and devices before granting access to communication services. All data transmitted over the network shall be encrypted using current industry standards. The system shall maintain a communication log for a minimum of ninety days, accessible only to authorised personnel.

The FRMCS infrastructure shall be designed to support a minimum of five simultaneous voice channels per train. The system shall provide guaranteed quality of service for safety-critical communications at all times. Network handover between base stations shall occur without any interruption to active communication sessions.

The system shall be capable of operating independently of public telecommunications networks in degraded mode. All onboard equipment shall comply with applicable European railway interoperability regulations. The system shall support remote configuration and firmware updates with zero service interruption. Infrastructure components shall have a minimum operational lifespan of fifteen years under normal railway operating conditions.